

Tax Shocks and County-Level Migration: Do High-Income Households Flee Higher Taxes?

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1 Introduction

We examine whether changes in state-level tax policy encourage high-income households to relocate to lower-tax jurisdictions.

Our primary data source is the IRS Statistics of Income (SOI) series, which publishes county-level income and migration data annually. The income data reported are adjusted gross income (AGI) and taxes paid by households aggregated to the level of county-income bracket-year. The migration data report the number of returns, number of individuals, and cumulative AGI for every county-to-county move in each year. Our sample spans **2011–2022**.

The unit of observation in our analysis is a **county-pair** $\times$ year. So for each origin county, we observe how many people moved to each destination county from a given year to the next, along with the income and tax characteristics of both counties.

To identify causal effects we will explore three state-level policy shocks with known effective dates:

1. **New Jersey (2018)**: top marginal rate raised from 8.97% to 10.75% on income above \$5M; threshold later lowered to \$1M in 2021.
2. **New York State (2021)**: top marginal rate raised from 6.85% to 9.85% on income above \$1M.
3. **Delaware (2018)**: estate tax repealed (previously 16% on inheritances above \$5.49M).

Our effective tax rate variable is computed as $(\text{Federal income tax} + \text{State \& local taxes paid})/\text{AGI}$, computed separately for high-income ($\text{AGI} > \$200\text{k}$) and low-income ($\text{AGI} \leq \200k) households in each county-year.

2 Data Construction

All the data we use is scraped from the IRS Statistics of Income (SOI) webpage. The two data series we use are county-level aggregates of household income and tax statistics during the period of 2011-2022. The IRS bins households by adjusted gross income (AGI), so information about households with AGI above \$200k is separate from those with AGI below \$200k (we combine all the other AGI bins and call these two groups high and low-income households). To protect privacy, the counties where the number of tax returns in an AGI bin is fewer than 20 have all the values for that county-year-AGI bin combination suppressed to zero.

Migration data comes from a separate series of publications also from SOI. It includes the number of tax returns which were filed in one county in year 1 and another in year 2, the fips codes of those two counties, the number of people represented in the tax returns, and the total AGI of these tax returns in the second year. We construct a panel of all the county migration pair-year combinations with various income and tax statistics about the two counties. Similar to the income and tax data, if the number of tax returns in a county pair-year combination is fewer than 10, the values for that observation are suppressed to -1.

Our analysis is not so concerned with the extensive margin (a county having any migrants), so we drop all observations where migration data are suppressed to -1. These observations are entirely very small counties, and we expect this decision not to bias any estimates. In the case where a county's income data is suppressed, the effective tax rates are NaN, and those counties are omitted from regressions.

3 Descriptive Analysis

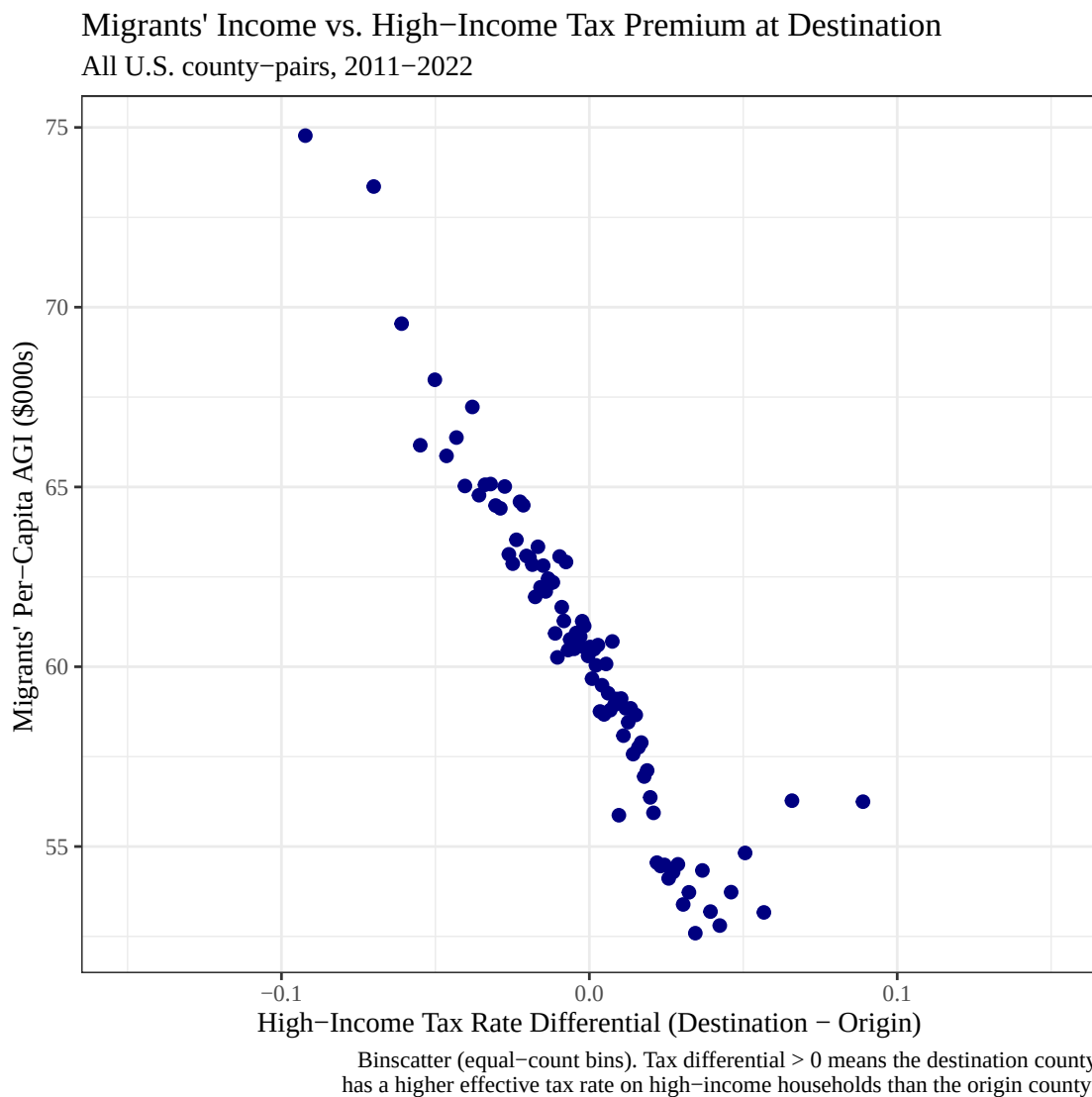
The figures below explore the unconditional relationships in our data before turning to causal identification. We ask two preliminary questions:

1. Across all county-pairs, do wealthier migrants tend to move toward lower-tax destinations?

- Does this pattern differ depending on whether we measure the tax rate faced by high-income or low-income households?

3.1 Migrants' Wealth vs. High-Income Tax Differential

Figure 1.



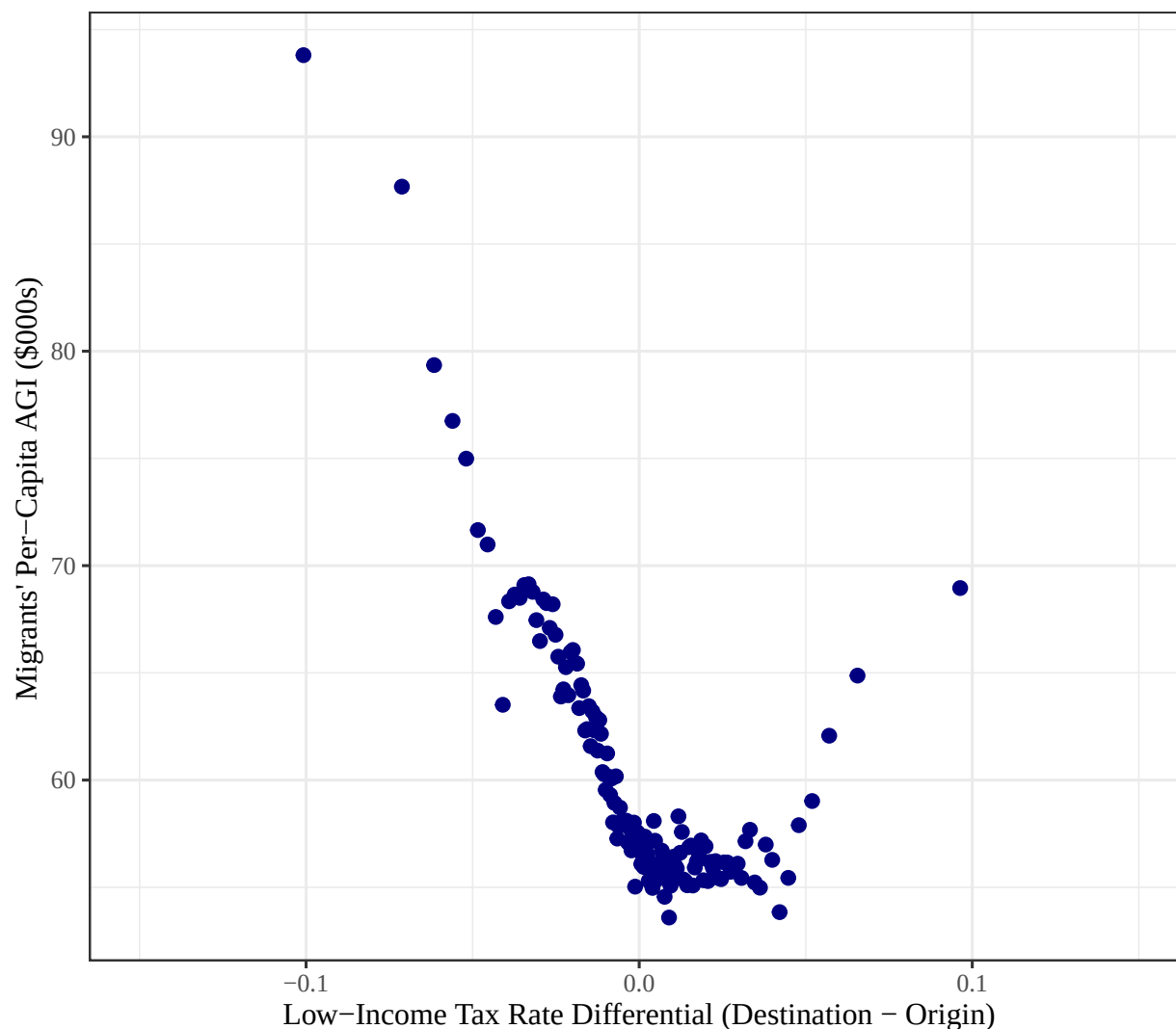
The downward slope of the figure indicates that the type of counties into which migrants have higher average incomes are counties where households with AGI above \$200k happen to pay lower tax rates than the county those migrants left. This is consistent with the basic logic of ‘rich people evading taxes.’

3.2 Migrants' Wealth vs. Low-Income Tax Differential

Figure 2.

Migrants' Income vs. Low-Income Tax Premium at Destination

All U.S. county-pairs, 2011–2022



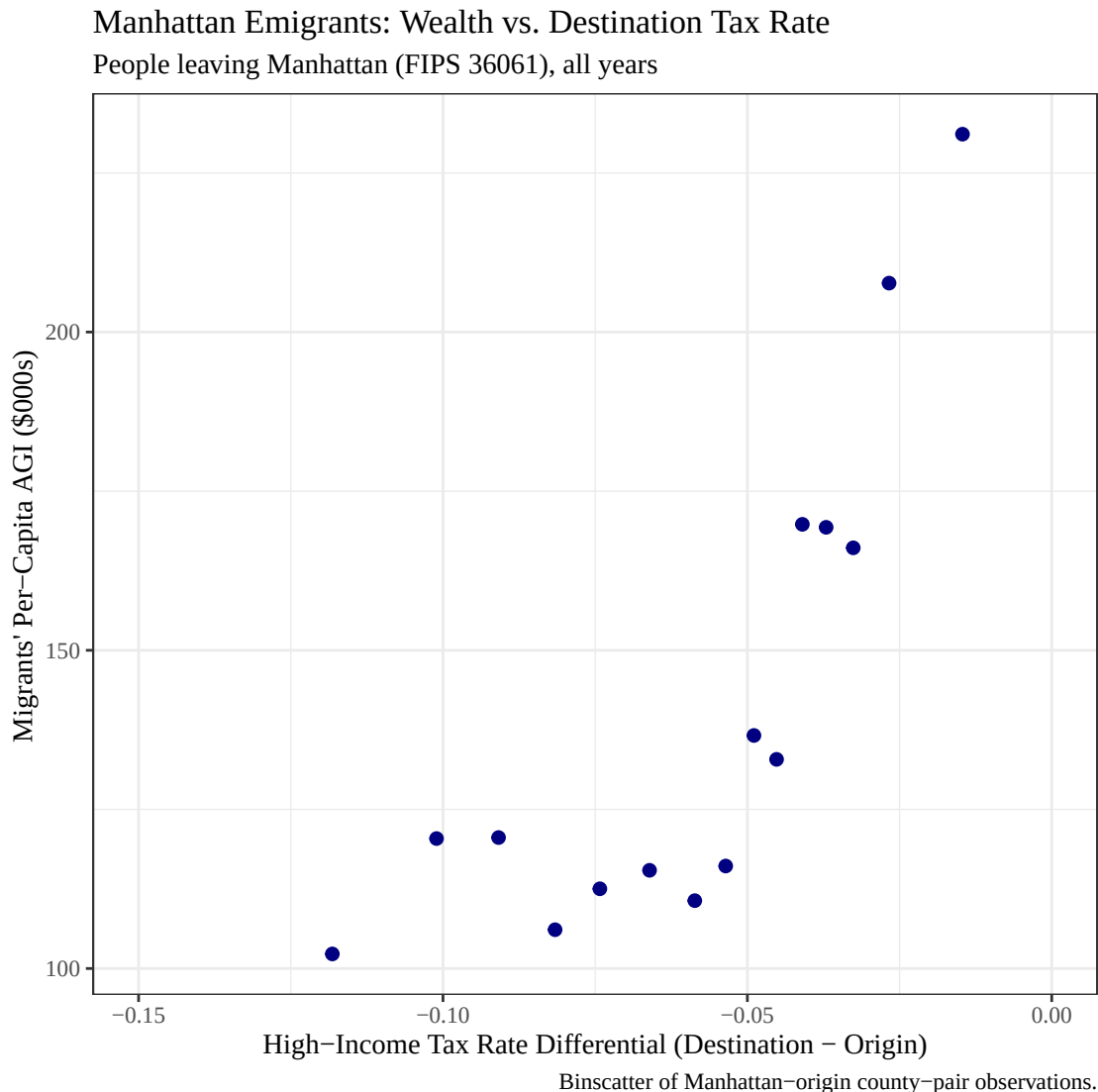
The pattern here is similar to Figure 1. Migrants' income is still negatively correlated with higher relative realized taxes in the destination county. This suggests positive correlation between taxes faced by high-income households and low-income (the bottom ~95%) households. We do not know how to account for the curve upwards near tax-premium = 5 pp.

4 Manhattan Case Study

To reduce cross-sectional noise we will look **Manhattan (New York County, FIPS 36061)**, one of the highest-tax, highest-income counties in the country. We ask: among the people leaving Manhattan NYC in any given year, does the destination county's effective high-income tax rate influence those migrants' average incomes?

A negative slope would suggest high-earning Manhattanites prefer moving to lower-tax destinations. A positive slope, which is what we observe, suggests something more nuanced: Manhattan-leavers tend to relocate to other high-tax metros (San Francisco, Boston, Seattle), rather than moving to minimize taxes.

Figure 3.



Across every year in our data, figure 3 bins observations by the size of the difference between high-income households' taxes in the destination county and Manhattan County in NYC. Interesting, the bins are such that every measurement of the Conditional Expectation Function has a negative tax differential value (which makes sense, because NYC is a high-tax place). Where the tax discount in the destination is less than 5 pp, there appears to be no relationship with the income of the people moving there. Counter-intuitively, and counter to the implications of figures 1 & 2, counties with higher realized taxes on high-income households (similar to NYC) attracted migrants with higher average incomes. This is explained by the allure of places like San Francisco outweighing the dreaded California tax code.

Something to notice about figure 3 is that every bin has a very high average income. Peering into the data some more, there are indeed some grotesquely rich people, featuring 166 households who filed their taxes in Manhattan in 2012 and in Albany county in 2013 who had an **average** income of ... **\$3,800,000**

From 2020 to 2021, there were 2500 households with an average income of \$1.2M moving from Manhattan to Miami-Dade County. Not even the Central Limit Theorem can hide the unbelievable right tail of the American income distribution.

5 A DD with New York State

In 2021, New York State raised its top marginal income tax rate from **6.85% to 9.85%** on income exceeding \$1 million. Our strategy capitalizes on the fact that some destination counties outside New York were already attracting rich New York migrants before the shock, while others were not. If the 2021 tax increase caused a behavioral response among high earners, the counties that were drawing wealthier New Yorkers in the pre-shock period should see a larger increase in inflows after the rate rose. If these counties also especially attracted the ultra-high-income households, these migrants' average incomes should increase too.

The figure below plots the **proportional change** in migration flow (average of 2021-2022 moves vs. the 2019 pre-shock baseline) against the **log per-capita AGI** of people who moved from each NY county to each destination in the pre-shock year. A positive slope would support the hypothesis that high-income New Yorkers were pushed out at an increased rate after the tax increase.

Note on year labels: Each `year` value in the data is the *starting* year of the IRS migration file, so `year == 2019` captures people who physically moved between filing their 2019 taxes and 2020 taxes, `year == 2020` captures moves between 2020 taxes and 2021 taxes (the immediate shock year), and `year == 2021` captures moves during 2022. The pre-shock baseline uses `year == 2019` (moves in 2020); the post-shock period averages `year %in% c(2020, 2021)` (moves in 2021 and 2022).

Figure 4.

NY Tax Shock: Change in Out-Migration by Destination's Previous Migrants' Income
 NY-origin county-pairs; avg. post-shock flows (2021–2022) vs. pre-shock baseline (2020)

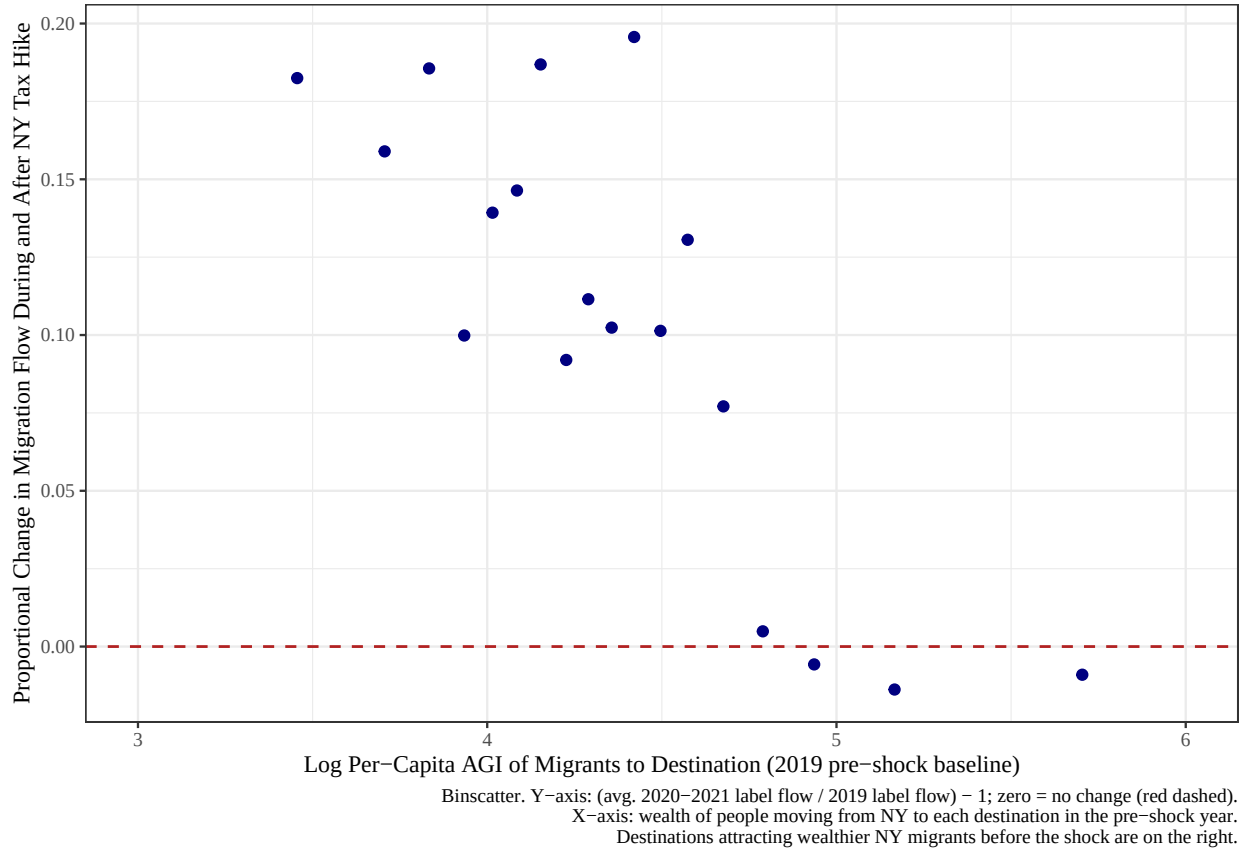


Figure 4 reports a binscatter plot of the percent increase in the number of households moving to each destination (vertical value) relative to the logarithm of the per-capita AGI of migrants to the same destination during 2020 (who left NY in 2019). The slope trends downward, but is very noisy for per-capita AGI less than \$116k (horizontal value = 4.75). Then the binscatter jumps down to the level indicating no relationship between pre-2020 migrant average income and a change in migration after the NY state tax increase.

6 Geographic Patterns: The NY 2021 Tax Shock on a Map

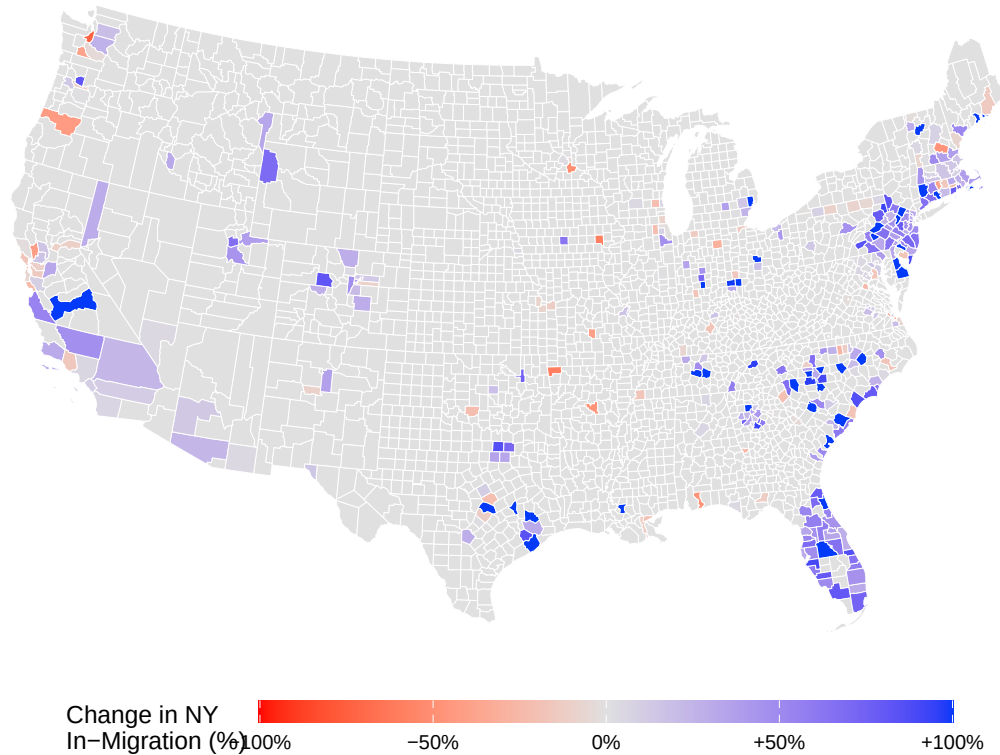
Figure 4 shows the *aggregate* relationship between NY out-migration and destination wealth. The maps below show the *geographic* version of the same question: which parts of the country absorbed New York migrants, and how did that pattern shift in the year taxes went up?

6.1 Where Did NY Migrants Go? (Before vs. After 2021)

Figure 5.

Where Did NY Migration Shift After the 2021 Tax Increase?

Percent change in arrivals from NY State: avg. 2021–2022 vs. avg. through 2020



The map complements the binscatter results. Blue counties absorbed more New York migrants after the 2021 rate increase; red counties absorbed fewer. The gains are concentrated in Florida (particularly Miami-Dade and Palm Beach counties), parts of the Carolina and Texas metros, and a handful of coastal destinations. Most of the country shows little change (white), meaning the shock did not produce a broad nationwide redistribution of New York migrants. The pattern is geographically clustered rather than uniformly driven by low taxes, pointing toward amenity sorting as a competing explanation alongside tax avoidance.

Note: An animated version of this map showing Manhattan out-migration year-by-year from 2011–2021 is available in `Updated Working Files/Analysis/Map Visualization.R`. Run that script to produce `manhattan_exodus_animated.gif`. The animation makes the 2021 shift visually apparent against the prior trend.

7 Regression Analysis

As mentioned above, New York State increase its top marginal income tax rate in 2021 from 6.85% to 9.85% for households with incomes exceeding \$1 million. To establish the type of analysis we intend to conduct,

we estimate the following equation:

$$\log(n) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat * Post)_{c,t} + \epsilon_{c,t}$$

where n is the number of tax returns moving from one county to another, $Treat_c$ is an indicator for if the county in year 1 was in New York, and $Post_t$ is an indicator for if the year of the observation was 2020 or 2021 (to capture people who moved anticipating the increase in taxes as well as people responding to the increase by leaving the following year). We also estimate the income composition model, where the only difference is that number of returns has been replaced by mean AGI of those tax returns:

$$\log\left(\frac{AGI}{n}\right) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat * Post)_{c,t} + \epsilon_{c,t}$$

Table 1

Table 1: NY 2021 DiD Estimates. Standard errors in parentheses.
*** $p < 0.001$.

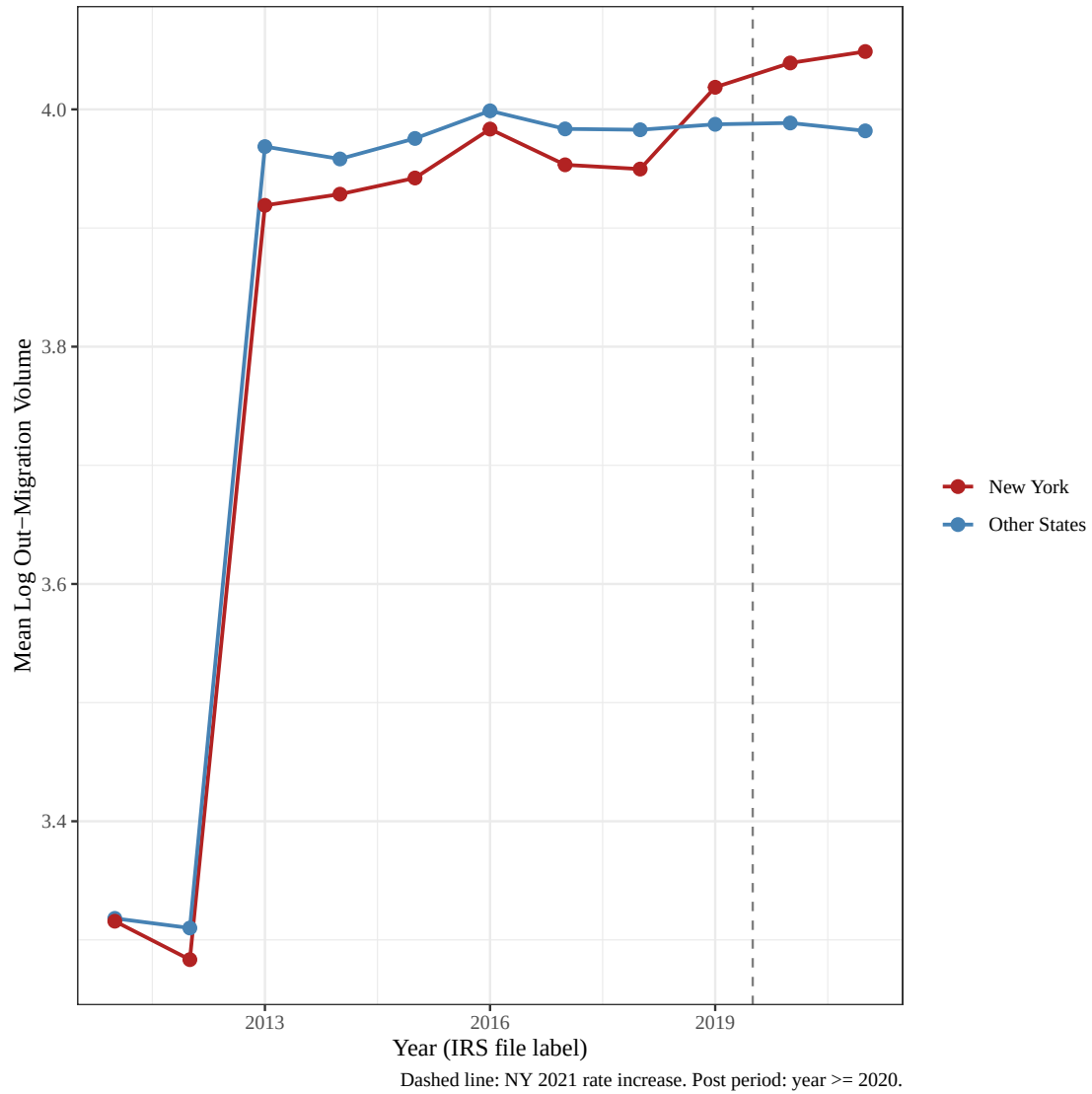
Coefficient	Volume: $\log(n)$	Income: $\log(AGI/n)$
β_1 (Treat)	-0.0089 (0.0075)	0.1871*** (0.0036)
β_2 (Post)	0.2366*** (0.0035)	0.2568*** (0.0017)
β_3 (Treat $\times$ Post)	0.0677*** (0.0176)	0.0693*** (0.0085)

There was a 23.7% increase in migration among counties outside of New York from the 2011-2020 average to the 2020-2022 average. The increase between the two periods for people leaving New York was 6.8 pp greater than the national average. As suggested by figure 1, the New York State tax increase did increase the number of New Yorkers fleeing the state.

The average income of people moving out of New York State during 2011-2022 was 19% higher than the people leaving anywhere else. There was a 26% increase in mean AGI of people leaving a county outside of NY from the 2011-2020 average to the 2020-2022 average. From the coefficient estimate of the interaction term, the increase in mean AGI of people leaving NY from the 2011-2020 average to the 2020-2022 average was 7% greater than the mean AGI of people leaving anywhere else. This corroborates the estimates in the first estimating equation, namely that the deep-pocketed New Yorkers **were encouraged to flee the tax hike**.

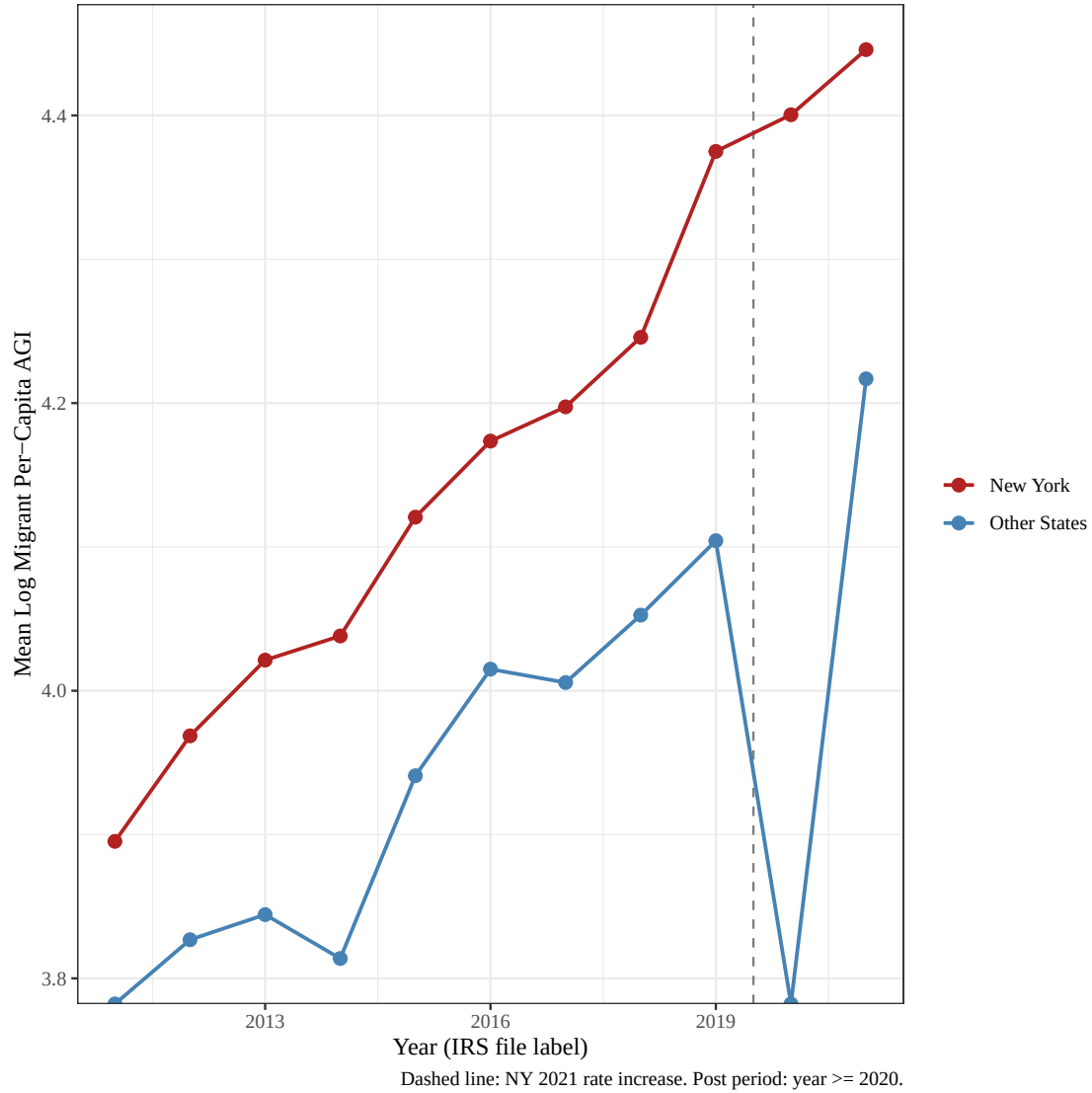
The figures below show mean log migration volume and mean log per-capita AGI over time for New York counties (treated) versus all other origin counties (control). The dashed line marks the start of the post period.

Figure 6.



The volume trends plot shows that New York out-migration moved broadly with the control group pre-tax increase, but rose sharply after it came into effect. This is consistent with the positive treatment by post coefficient.

Figure 7.



The income trends plot shows that New York out-migration have higher average per-capita AGI that migrants from other states throughout the period. Post 2021, this gap appears to widen.

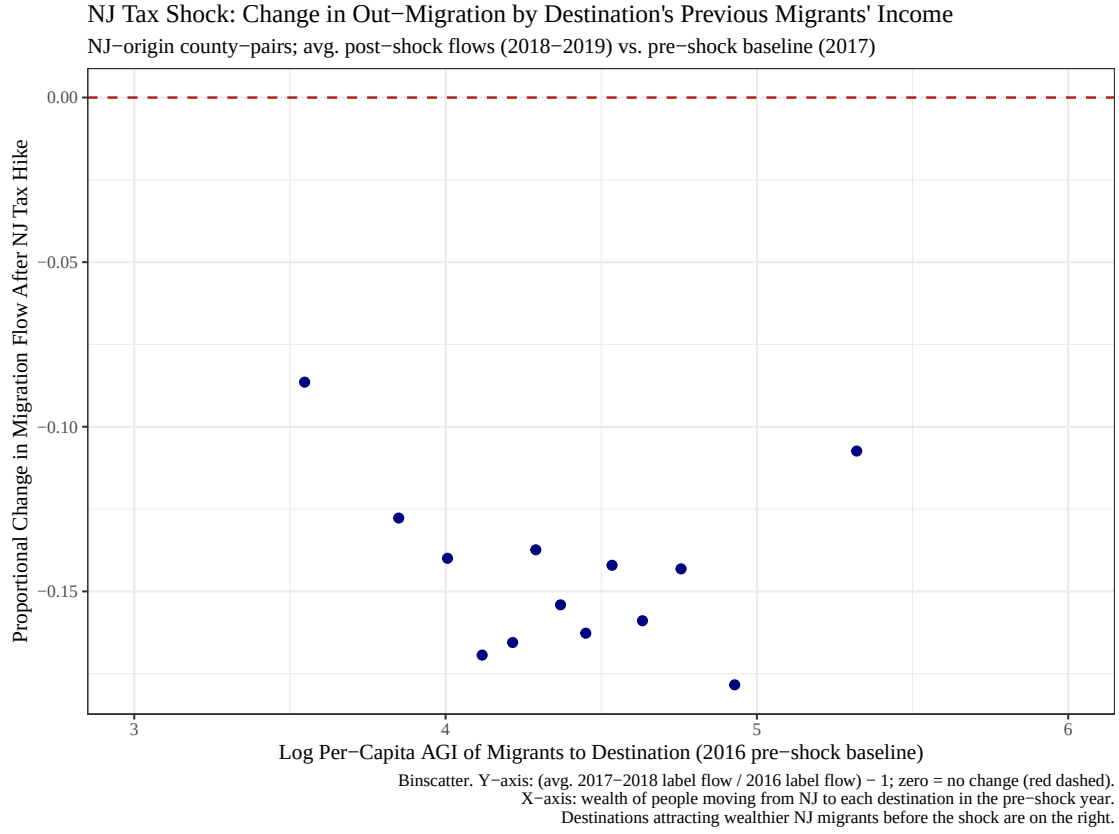
7.1 Disclaimer!!!

The policy change happened at the same time as a pandemic. It is impossible to distinguish between the effect of the tax change and the effect of the pandemic in NYC. This seriously confounds our analysis.

8 New Jersey 2018 Tax Hike

New Jersey raised its top marginal income tax rate in 2018, increasing the top rate from 8.97% to 10.75% for the highest income households. This policy change provides a useful comparison to the New York tax shock as there is more post-event data (and well before the pandemic), allowing us to look at longer-run migration patterns. For this regression, the treatment is county pair observations where the county of origin is New Jersey.

Figure 8.



In figure 8, there seems to be little relationship between pre-policy change migrant incomes and the subsequent change in the number of households leaving NJ.

We estimate the effect on out-migration volume:

$$\log(n) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

where $Treat_c = 1$ if the origin county is in New Jersey and $Post_t = 1$ for years 2017–2018 (moves during 2018–2019). We also made a model that looks at the income composition of movers:

$$\log\left(\frac{AGI}{n}\right) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

Table 2

Table 2: NJ 2018 DiD Estimates. Standard errors in parentheses.

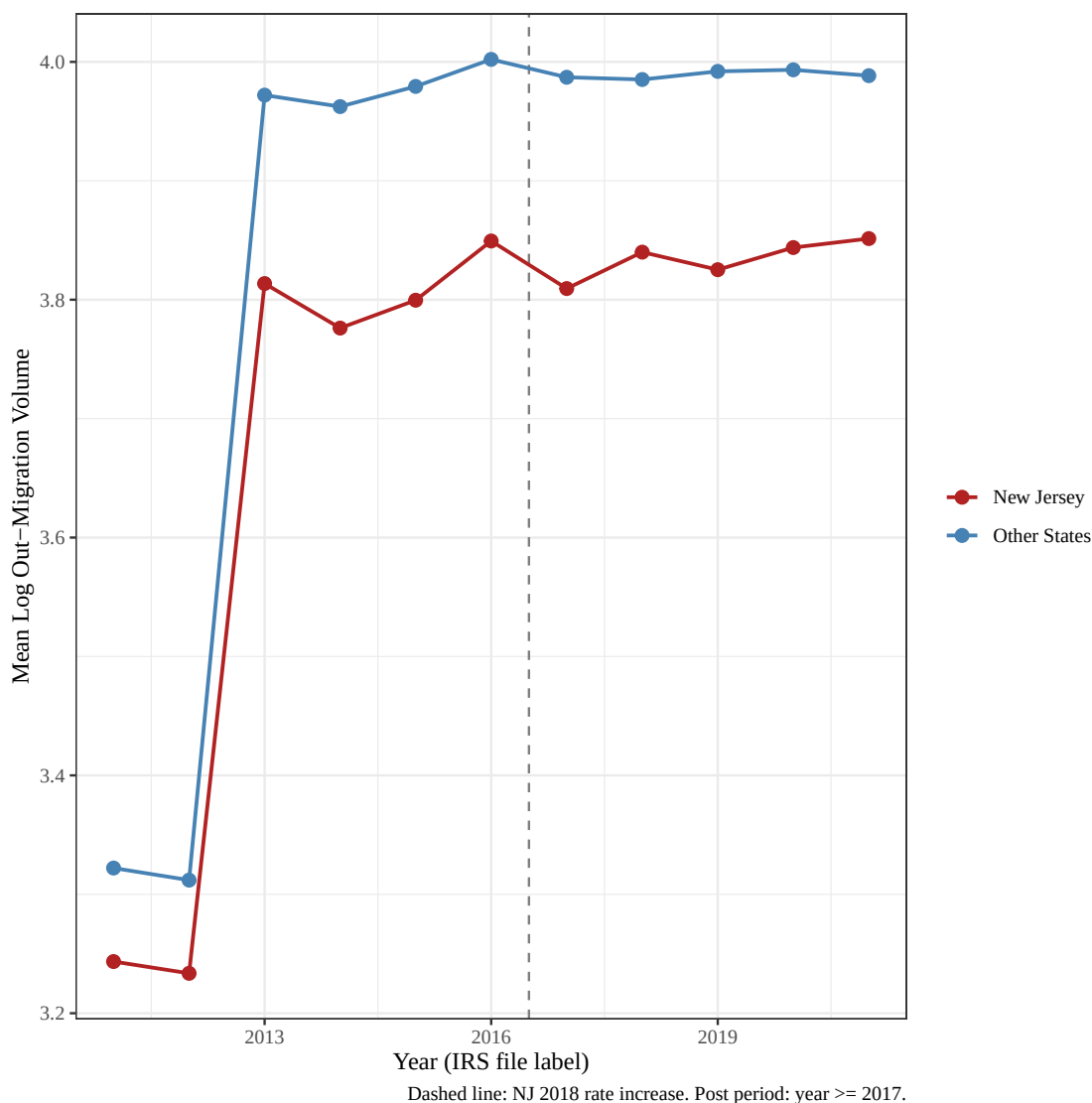
Coefficient	Volume: $\log(n)$	Income: $\log(AGI/n)$
β_1 (Treat)	-0.1221*** (0.0093)	0.4037*** (0.0045)
β_2 (Post)	0.2281*** (0.0036)	0.0876*** (0.0017)
β_3 (Treat $\times$ Post)	-0.0398. (0.0233)	-0.0038 (0.0114)

The New Jersey estimates do not show strong evidence of a large response to the 2018 tax increase. The volume model shows that tax filer flows that originated in New Jersey were lower than flows from other

points of origin on average, and the post-period coefficient indicates that migration volume increased in general after 2017 among the control group. However, the interaction coefficient is negative and only slightly significant, which suggests the New Jersey out-migration was slightly less than other states after the tax hike. The income model shows that migrants leaving New Jersey had substantially higher mean AGI than migrants leaving other states (40% higher), but the treatment by post coefficient is very small and statistically insignificant. Altogether, this means that the tax increase did not clearly changes the income positions of people leaving New Jersey.

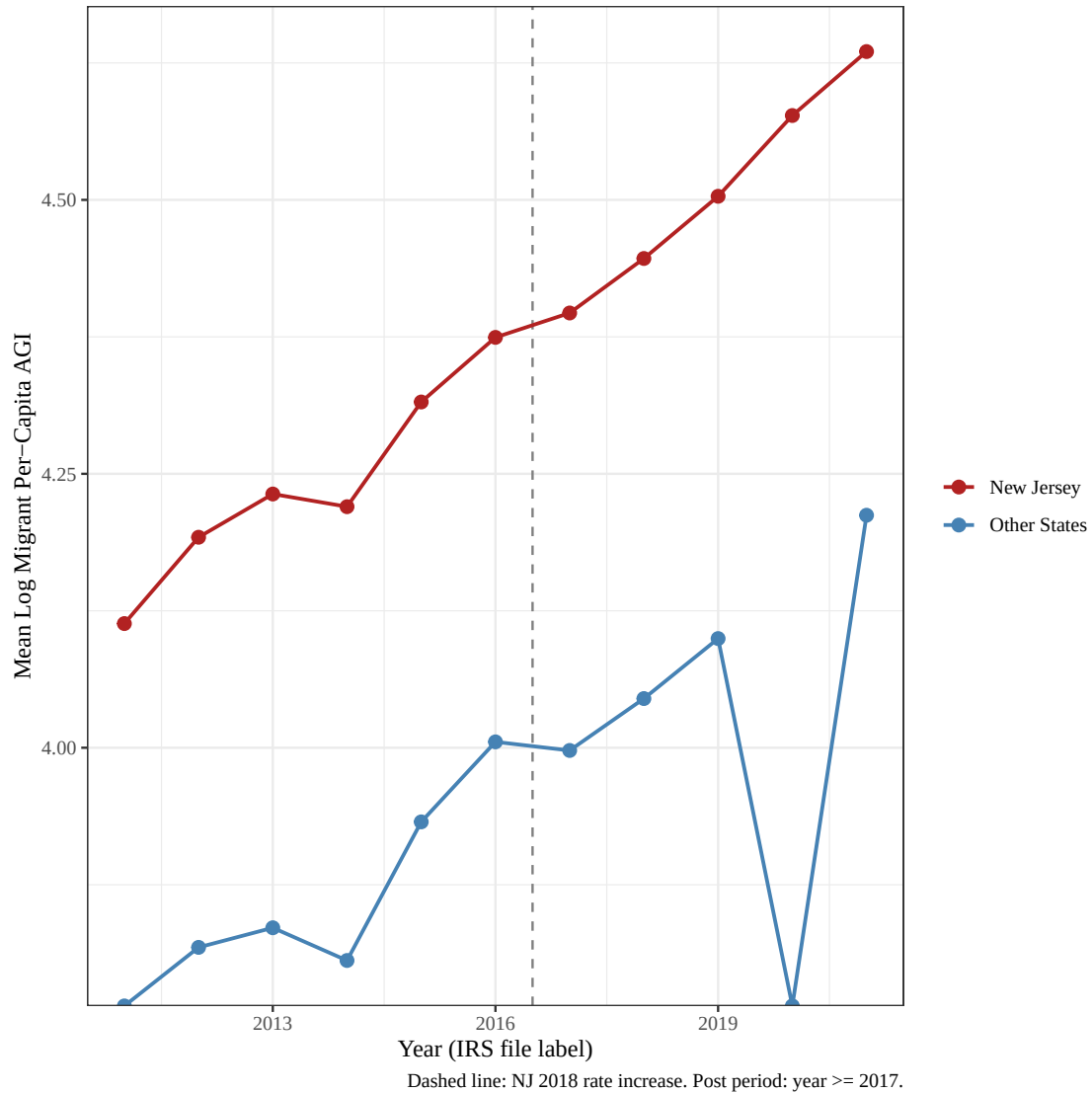
The figures below show mean log migration volume and mean log per-capita AGI over time for New Jersey counties (treated) versus all other origin counties (control). The dashed line marks the start of the post period.

Figure 9.



The volume trends plot shows that New Jersey out-migration moves broadly with the control group, and no clear post 2017 surge that would indicate that the change in tax policy resulted in a state-wide exodus.

Figure 10.

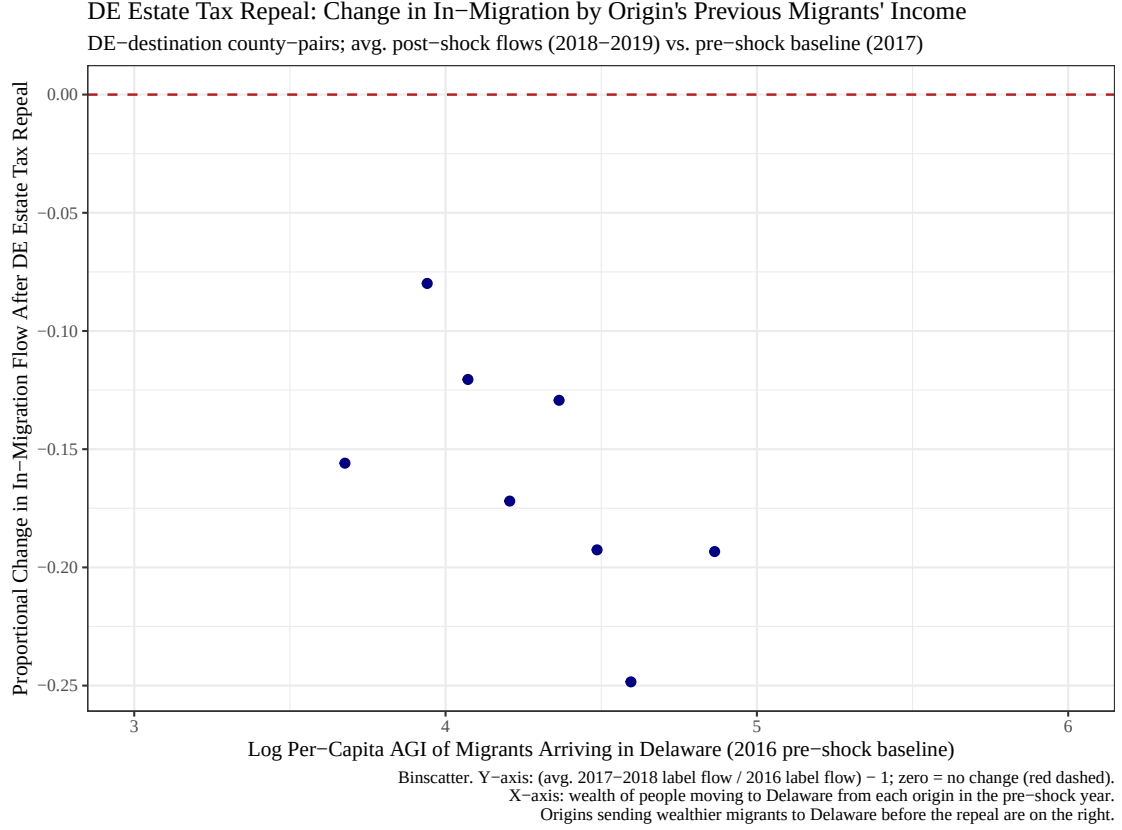


The income trends plot shows that New Jersey out-migrants had higher average per-capita AGI than migrants from other states. However, again there is not a clear break after the 2017 tax policy change that would suggest a change in income composition of movers.

9 Delaware 2018 Estate Tax Repeal

Delaware repealed its estate tax in 2018, which was previously only pertinent to very large estates. Compared to the previous analyses, this policy is a tax cut rather than an increase, so the expected behavior would be in the opposite direction. Instead of looking at whether individuals are leaving a high tax state, were looking at whether the repeal made Delaware more attractive (on top of other effects). Because of this, we decided to put the treatment on the destinations side rather than origin.

Figure 11.



From figure 11, it appears that counties from which migrants had higher average incomes had the largest decline in the number of people moving into Delaware after Delaware repealed their estate tax.

We estimate the effect on in-migration volume to Delaware:

$$\log(n) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

where $Treat_c = 1$ if the destination county is in Delaware and $Post_t = 1$ for years 2017–2018 (moves during 2018–2019). We also estimate the income composition model:

$$\log\left(\frac{AGI}{n}\right) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

Table 3: DE 2018 DiD Estimates. Standard errors in parentheses.

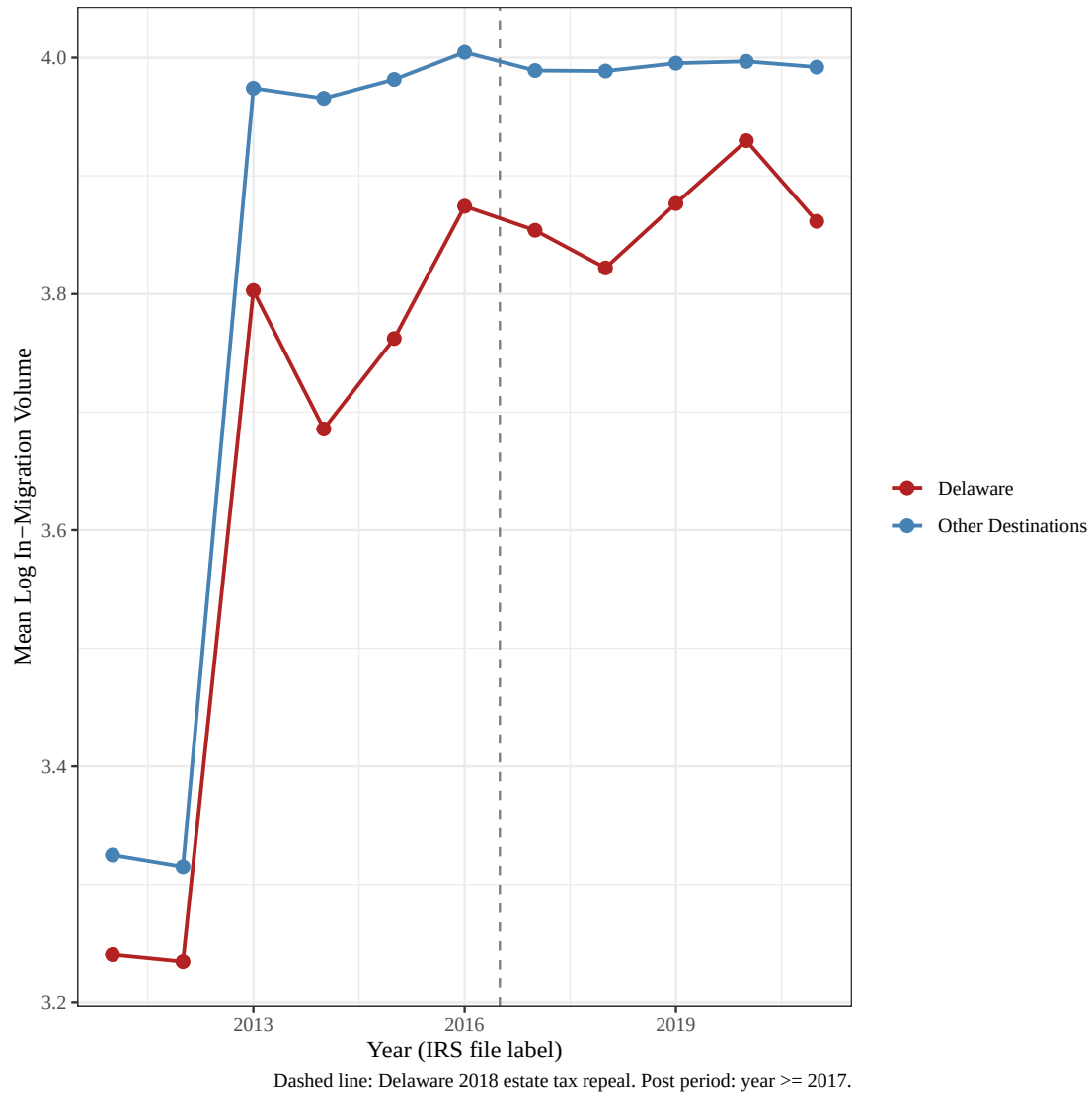
Coefficient	Volume: $\log(n)$	Income: $\log(AGI/n)$
β_1 (Treat)	-0.1148*** (0.0236)	0.2558*** (0.0116)
β_2 (Post)	0.2276*** (0.0035)	0.0876*** (0.0017)
β_3 (Treat $\times$ Post)	-0.0365 (0.0591)	0.0215 (0.0290)

Again, the Delaware regression results provide evidence against the notion that the estate tax repeal would produce a large increase in migration into Delaware. The volume model shows that Delaware destination flows were lower than other flows on average, while the post-period coefficient shows that migration volume rose across the board after 2017 in the control group. The interaction coefficient is negative but not statistically

significant, meaning that we cannot conclude that Delaware received either more or fewer migrants compared to other destinations post-repeal. The income model shows that migrants moving to Delaware had higher average AGI than migrants moving to other destinations, however, the interaction term is small, positive, and statistically insignificant. Overall, the repeal does not appear to have produced a clear increase in either the number or average income of migrants moving to Delaware.

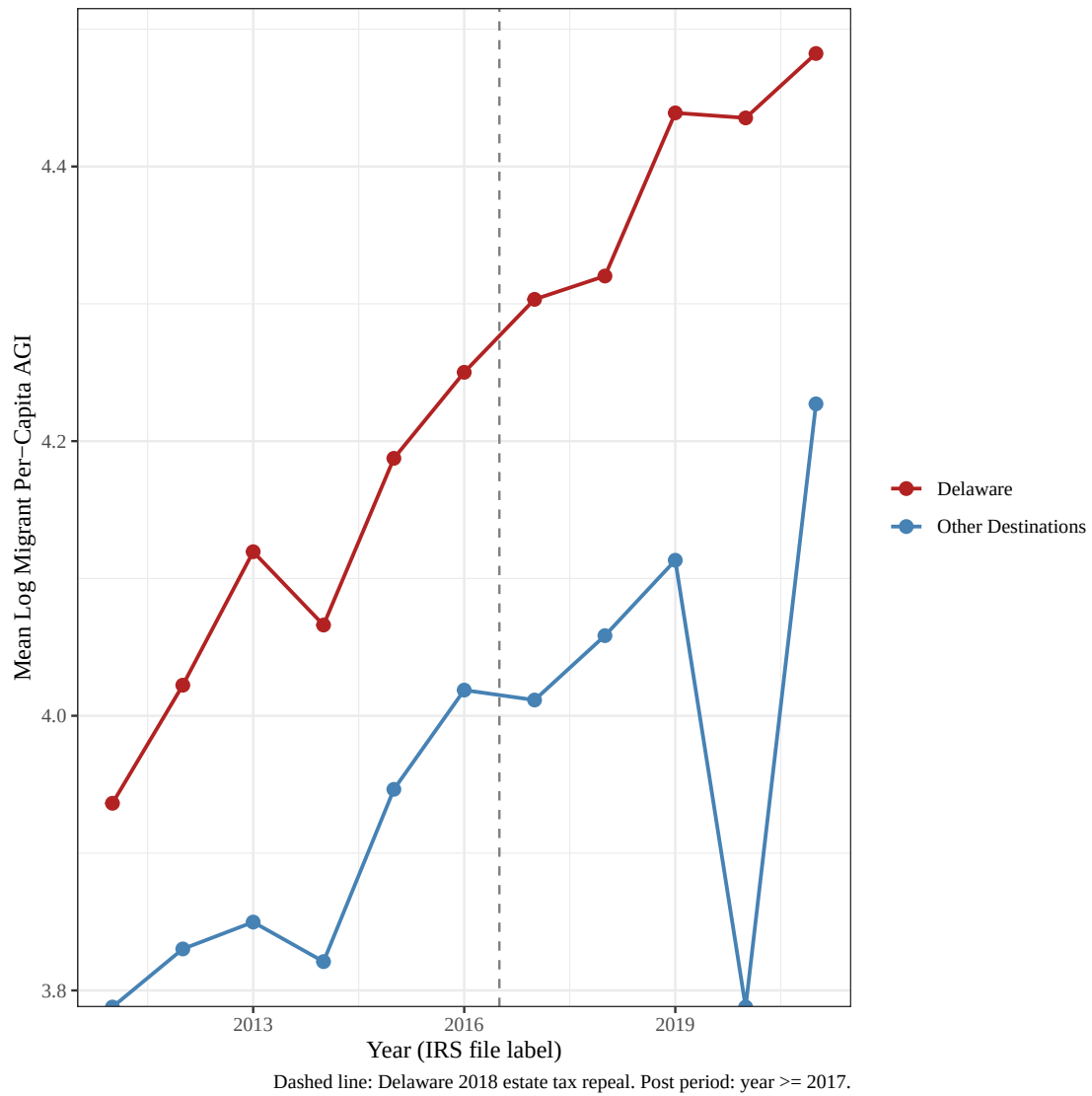
The figures below show mean log in-migration volume and mean log per-capita AGI over time for flows into Delaware counties (treated) versus flows into all other destinations (control). The dashed line marks the start of the post period.

Figure 12.



The volume trends plot shows no clear post 2017 increase in Delaware in-migration relative to other destinations, suggesting that the tax repeal did not visibly change Delawares migration volume trend.

Figure 13.



The income trends plot shows that migrants moving into Delaware had relatively high per-capita AGI, but the post repeal pattern does not show a visible break from the control group's trend.

10 Causality Discussion

Berg needs to include the results and figures from the Regressions 2 script. Discuss how the DDs for NJ and Delaware disagree with the DD for New York (probably because of the pandemic). Also discuss what the estimates from his TWFEs say.

That is all to give context for the limitations of our study. Discuss what would need to happen next to make a proper causal inference paper. Contextualize our estimates.

11 Reproducibility

This document was produced using **R** and **RMarkdown**. To reproduce it on any machine:

1. **Clone the repository** from GitHub.
2. **Open the project** by double-clicking `Data-Analysis-Project.Rproj` in RStudio. This sets the working directory to the project root automatically, do not open the `.Rmd` file directly from File Explorer/Finder, as paths will break.
3. **Knit the document**: open `Project Write-Ups/Stage_Two.Rmd` and click *Knit*, or run:

```
rmarkdown::render("Project Write-Ups/Stage_Two.Rmd")
```

4. **Packages are installed automatically** the first time you knit. No manual `install.packages()` calls are needed.
5. **The cleaned data file** (`Updated Working Files/Clean Data/cleanMigrationData.RData`) is committed to the repository. As long as this file is present, no scraping or data-cleaning scripts need to be re-run.

Note on the map section (Section 5): The geographic figure downloads U.S. county shapefiles from the U.S. Census Bureau via the `tigris` package. This requires an internet connection on the first knit only; all subsequent knits use a local cache.

If `cleanMigrationData.RData` has been deleted, run the scripts in this order before knitting:

```
# Step 1 - only needed if Temp Data/ files were also deleted (requires internet)
source("Data Wrangling/Data Scraping.R")

# Step 2 - regenerates cleanMigrationData.RData from Temp Data/ files
source("Data Wrangling/Cleaning and Merging.R")
```

Session info at time of submission:

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.0
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/Denver
## tzcode source: internal
```

```
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] here_1.0.2      scales_1.4.0    sf_1.0-23      tigris_2.2.1
## [5] fixest_0.13.2   binsreg_1.1     lubridate_1.9.4 forcats_1.0.1
## [9] stringr_1.5.2   dplyr_1.1.4     purrr_1.2.1    readr_2.1.6
## [13] tidyr_1.3.1     tibble_3.3.0    ggplot2_4.0.0  tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      xfun_0.53        lattice_0.22-7
## [4] tzdb_0.5.0        numDeriv_2016.8-1.1 vctrs_0.7.2
## [7] tools_4.5.1       generics_0.1.4    sandwich_3.1-1
## [10] proxy_0.4-27      pkgconfig_2.0.3   Matrix_1.7-3
## [13] KernSmooth_2.23-26 RColorBrewer_1.1-3 S7_0.2.0
## [16] stringmagic_1.2.0 uuid_1.2-2        lifecycle_1.0.4
## [19] compiler_4.5.1    farver_2.1.2      MatrixModels_0.5-4
## [22] SparseM_1.84-2    quantreg_6.1      htmltools_0.5.8.1
## [25] class_7.3-23      yaml_2.3.10       Formula_1.2-5
## [28] pillar_1.11.1     MASS_7.3-65       classInt_0.4-11
## [31] nlme_3.1-168      tidyselect_1.2.1  digest_0.6.37
## [34] stringi_1.8.7     splines_4.5.1     rprojroot_2.1.1
## [37] fastmap_1.2.0     grid_4.5.1        cli_3.6.5
## [40] magrittr_2.0.4    survival_3.8-3    e1071_1.7-16
## [43] withr_3.0.2       dreamerr_1.5.0     rappdirs_0.3.3
## [46] timechange_0.3.0  rmarkdown_2.30    httr_1.4.7
## [49] matrixStats_1.5.0 hms_1.1.4         zoo_1.8-14
## [52] evaluate_1.0.5    knitr_1.50        rlang_1.1.7
## [55] Rcpp_1.1.1        glue_1.8.0        DBI_1.2.3
## [58] rstudioapi_0.18.0 R6_2.6.1          units_1.0-0
```